STACKS, DISCRIMINANTS: LEMMA-DIFFERENT-RAMIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1906-1912 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1914-1939, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_discriminant.json (pb=1351, stmt 1906-1912, proof 1914-1939).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BIL: sources/stacks-discriminant/pinned-original-source.tex, source lines 1939–1939. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BW7: sources/stacks-discriminant/pinned-original-source.tex, source lines 1813–1827. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BJF: sources/stacks-discriminant/pinned-original-source.tex, source lines 756–759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BW8: sources/stacks-discriminant/pinned-original-source.tex, source lines 1863–1867. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0476: sources/stacks-discriminant/pinned-original-source.tex, source lines 1939–1939. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GH: sources/stacks-discriminant/pinned-original-source.tex, source lines 1939–1939. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : Y \to X$ be a flat quasi-finite morphism of Noetherian schemes. The closed subscheme $R \subset Y$ defined by the different $\mathfrak{D}_f$ is exactly the set of points where f is not étale (equivalently not unramified).*

Proof. Since f is of finite presentation and flat, we see that it is étale at a point if and only if it is unramified at that point. Moreover, the formation of the locus of ramified points commutes with base change. See Morphisms, Section 02GH and especially Morphisms, Lemma 0476. By Lemma 0BW7 the formation of R commutes set theoretically with base change. Hence it suffices to prove the lemma when X is the spectrum of a field. On the other hand, the construction of $(\omega_{Y/X}, \tau_{Y/X})$ is local on Y. Since Y is a finite discrete space (being quasi-finite over a field), we may assume Y has a unique point.

Say $X = \mathrm{Spec}(k)$ and $Y = \mathrm{Spec}(B)$ where k is a field and B is a finite local k-algebra. If $Y \to X$ is étale, then B is a finite separable extension of k, and the trace element $\mathrm{Trace}_{B/k}$ is a basis element of $\omega_{B/k}$ by Fields, Lemma 0BIL. Thus $\mathfrak{D}_{B/k} = B$ in this case. Conversely, if $\mathfrak{D}_{B/k} = B$, then we see from Lemma 0BW8 and the fact that the norm of 1 equals 1 that the discriminant is empty. Hence $Y \to X$ is étale by Lemma 0BJF. □